

6.131 Final Project Report

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1 Introduction

The human eye can only capture a limited number of images per second, at a maximum frequency of 50-90 Hz.¹ As a result, if a light is flashed at a slightly higher frequency, usually between 80 and 2000 Hz, then even though the flash itself might be slightly too fast for human eyes to perceive, it can cause the eye to only catch scattered glimpses of an object in motion, making continuous motion appear to be static, slowed down, or even reversed.² This type of optical illusion, where a flashing light is used to change how motion is perceived by an observer, is called the stroboscopic effect.

This project is an LED stroboscope that allows for the demonstration and testing of the stroboscopic effect. More specifically, this project uses an LED array flashing at an adjustable rate to show how a wheel spinning at a consistent speed can appear to slow down, speed up, reverse, and stop moving entirely depending on the frequency of the flashing light.

2 Hardware System Design

There are two main systems involved in this project: a wheel that spins at a constant speed, and an LED array that flashes at an adjustable frequency. Together, the rotating wheel and flashing LED array demonstrate the stroboscopic effect.

2.1 Wheel System Design (Buck Converter With Feedback)

The rotating wheel is controlled by a 5-volt DC motor. This DC motor is powered by a buck converter that steps down the 12-volt input lab power supply to 5 volts, using feedback control in order to keep the voltage consistent. The schematics for this system are shown in Figures 1 and 2.

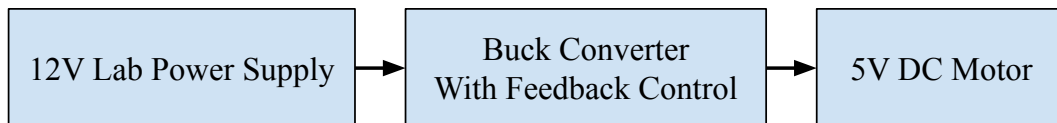


Figure 1: A block diagram of the wheel system design.

¹Davis, J., Hsieh, YH. & Lee, HC. Humans perceive flicker artifacts at 500 Hz. Sci Rep 5, 7861 (2015). <https://doi.org/10.1038/srep07861>

²CIE, “Visual aspects of time-modulated lighting systems—definitions and measurement models,” CIE TN 006:2016 (2016).

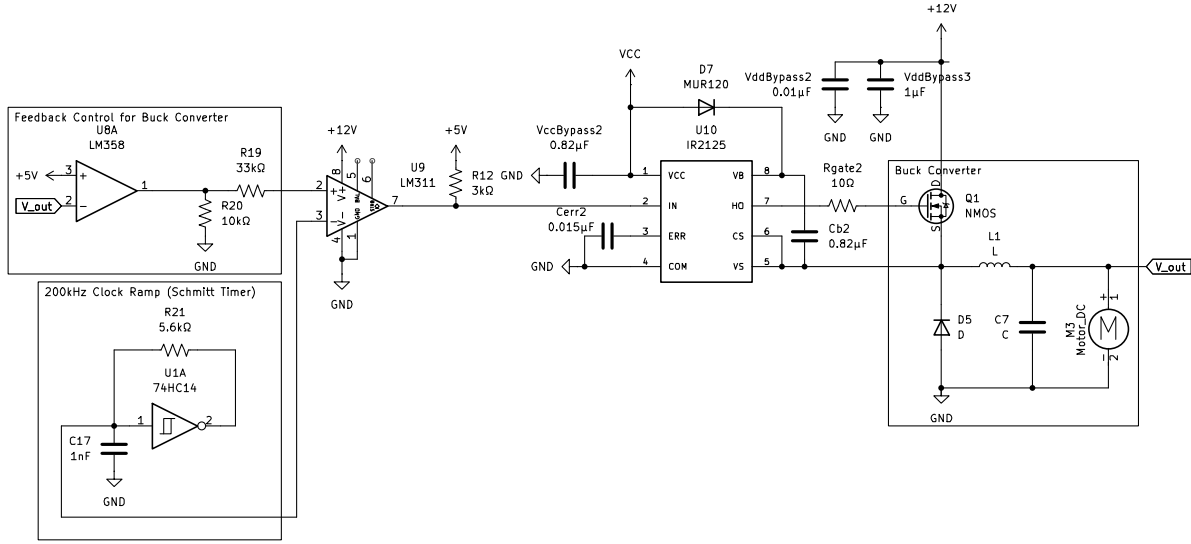


Figure 2: A detailed KiCad schematic of the wheel system design.

Buck Converter With Feedback Constraints:

- $V_{in} = 12V$
- $V_{out} = 5V$
- $f_{sw} = 200 \text{ kHz}$
- $f_r < 10\%$ of f_{sw}
- With a 12V input, ensure that the buck converter stays in CCM at a power corresponding to the motor being run at 5V (i.e. if estimating a motor resistance of 15Ω , ensure buck converter is in CCM for a minimum power of $\frac{5^2}{15} \approx 1.67W$)
- $\Delta v_{spec} < 0.25V$ (5% of 5V)

Based on the CCM constraint, $L_{min} = \frac{V_{out}^2 DT}{2P_{min}} = \frac{(5^2)(1-\frac{5}{12})(\frac{1}{200000})}{2(1.67)} = 0.000021875... \rightarrow L_{min} \approx 21.9\mu H$

Based on the Δv_{spec} constraint, $C_{min} = \frac{V_{out}(1-D)T^2}{8L\Delta v_{spec}} = \frac{5(\frac{5}{12})(\frac{1}{200000})^2}{8(21.9 \cdot 10^{-6})(.25)} = 0.000001189... \rightarrow C_{min} \approx 1.2\mu F$

Based on the f_r constraint, ensure that $f_r = \frac{1}{2\pi\sqrt{LC}} < 20,000Hz$. For current L_{min} , C_{min} , $f_r \approx 31,046Hz$.

Final Component Values:

$L = 100\mu H$, T90-52 iron powder core with $A_L = 64$ and approximately $N = 40$ turns of 24 AWG wire

$C = 220\mu F$, electrolytic capacitor in parallel with a $.82\mu F$ metal film capacitor

$f_r \approx 1,000 \text{ Hz}$

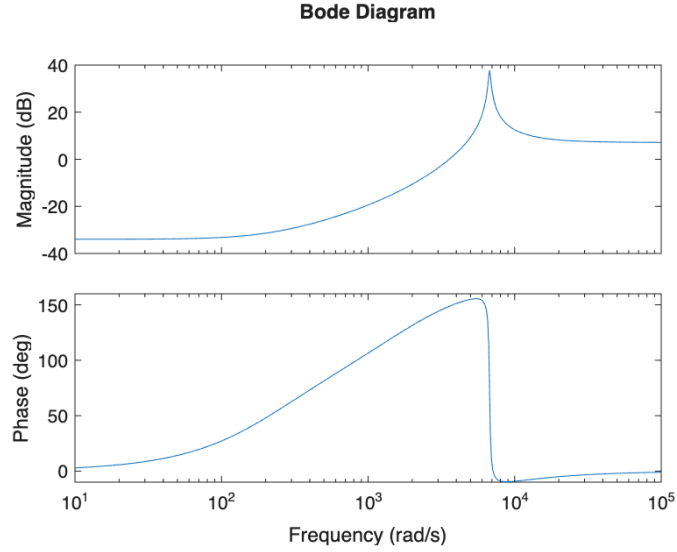


Figure 3: A Bode plot modeling the output impedance for L and C values (including parasitic resistances).

2.2 LED Array System Design (DCM Boost Converter)

The LED array is controlled by a DCM boost converter that steps up the 5-volt input power supply to a nominal 12-volt output. Having the boost converter be in DCM allows for better efficiency, since the current in DCM goes to 0A at the beginning of every cycle, mimicking the desired behavior of the flashing LED array and therefore reducing instability. Additionally, two oscillator circuits (one at a higher frequency, and one at a lower frequency) are combined together to allow the LED array to flash at an adjustable frequency ranging from around 10 Hz to 2000 Hz, while still ensuring that the boost converter can operate smoothly. The schematics for this system are shown in Figures 4 and 5.

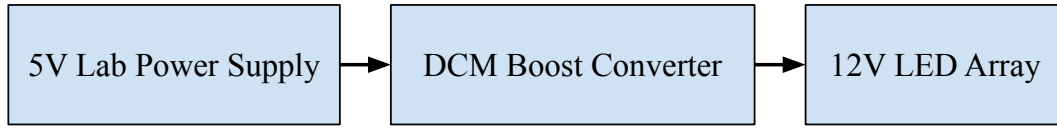


Figure 4: A block diagram of the wheel system design.

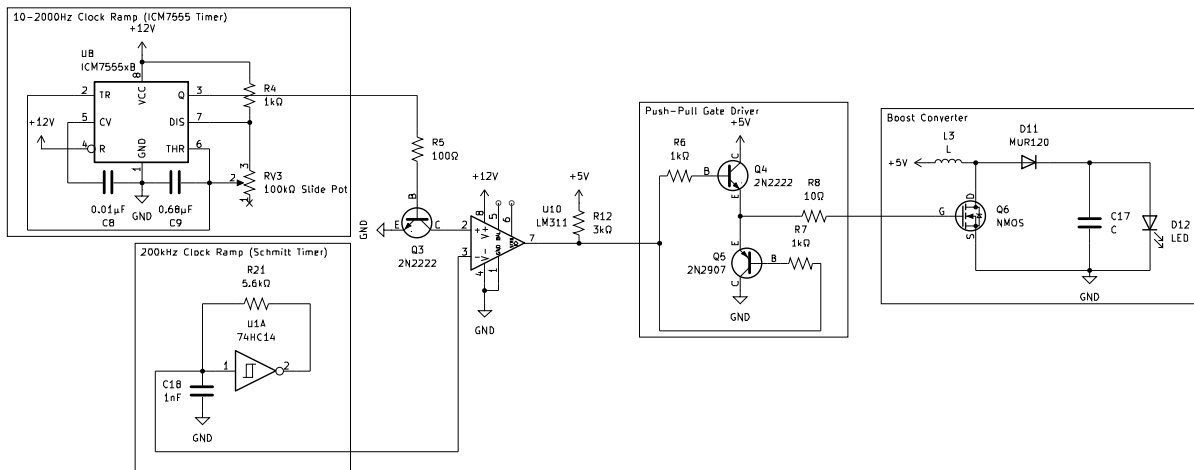


Figure 5: A block diagram of the wheel system design.

DCM Boost Converter Constraints:

- $V_{in} = 5V$
- $V_{out} = 12V$
- $f_{sw} = 200 \text{ kHz}$
- With a 5V input, ensure that the boost converter always stays in DCM for the 1.6-watt LED array.
- $\Delta v_{spec} < 0.6V$ (5% of 12V)

Note that at the edge of DCM, $\frac{I_{load}}{1-D} = \frac{V_{in}DT}{2L}$, meaning $L = \frac{V_{in}^2DT}{2P_{out}}$ and $C = \frac{P_{out}DT}{V_{out}\Delta v_{spec}}$.

Based on the DCM constraint, $L_{max} = \frac{V_{in}^2DT}{2P_{out}} = \frac{(5^2)(1-\frac{5}{12})(\frac{1}{200000})}{2(1.6)} = 0.00002278... \rightarrow L_{max} \approx 22.7\mu H$

Based on the Δv_{spec} constraint, $C \approx \frac{P_{out}DT}{V_{out}\Delta v_{spec}} = \frac{(1.6)(\frac{7}{12})(\frac{1}{200000})}{(12)(0.6)} = 0.000000648... \rightarrow C \approx 650nF$

Final Component Values:

$L = 20\mu H$, T90-52 iron powder core with $A_L = 64$ and approximately $N = 17$ turns of 24 AWG wire

$C = 470nF$, metal film capacitor

3 Buck Converter With Feedback

The buck converter with feedback can be thought of as three smaller components: the 200 kHz clock ramp, the buck converter itself, and the feedback control for the buck converter.

First is the 200kHz clock ramp. Originally, the 200 kHz sawtooth wave was going to be generated by an LM555 in astable mode, but the frequency proved too high for the component to properly reach.

For example, Figure 6 shows the output of pin 6 of the LM555 when using $C = 1nF$ and $R_A = R_B = 1.6k\Omega$, which should have theoretically generated a frequency of $\approx 300kHz$. Instead, the sawtooth wave generated was broken, and at a much lower frequency than expected.

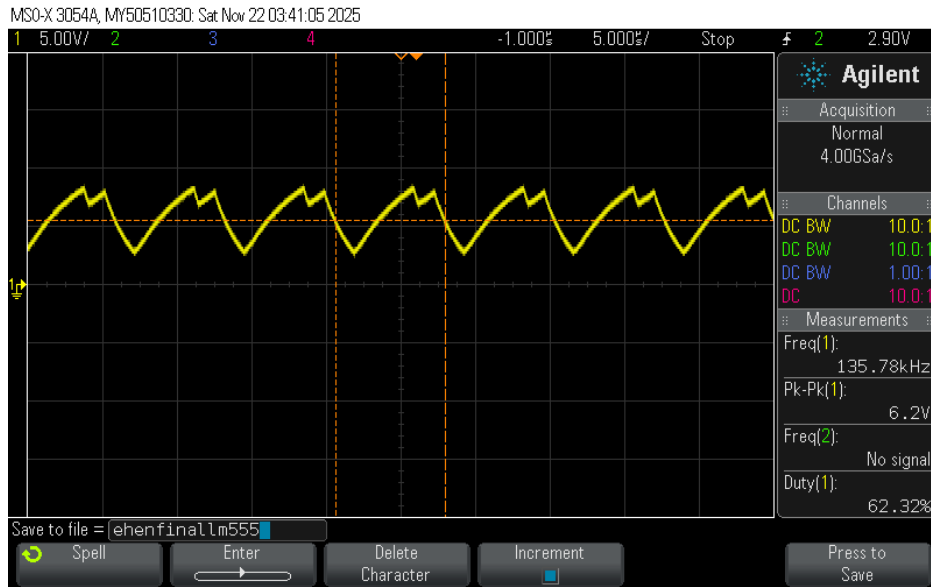


Figure 6: The yellow line indicates the output of pin 6 of the LM555.

As a result, the sawtooth wave is instead generated using a Schmitt trigger. A frequency of roughly 200kHz is obtained by using $C = 1nF$ and $R = 5.6k\Omega$, as shown in Figure 7.

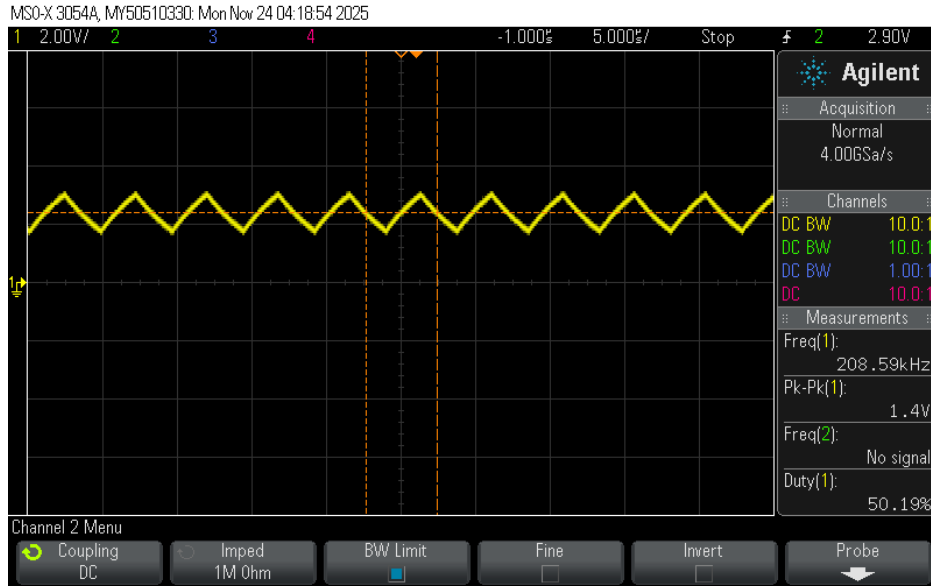


Figure 7: The yellow line indicates the output of the Schmitt trigger.

The next part of this module is the buck converter itself, which was built using the previously calculated values for L and C . The output of the buck converter (without feedback) can be seen in Figure 8, when using a 47Ω power resistor as a load.

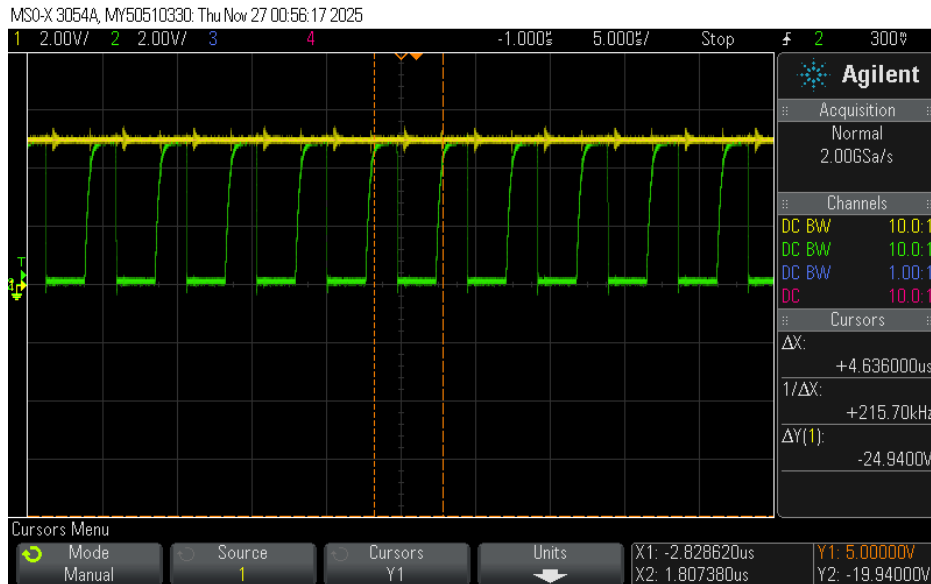


Figure 8: The yellow line is the output of the buck converter. The green line is the output of the LM311.

Finally, feedback was added to the buck converter in order to ensure it stayed at a constant 5V, regardless of the load (a property that would reduce noise in the buck converter output and allow for the easy testing of multiple motors).

In order to get proper feedback, a 5-volt reference voltage was needed. This was obtained using an LM7805, which produces a stable 5V voltage as shown in Figure 9. Future iterations could potentially use a specialized reference voltage chip, but this works well for the intended purpose.

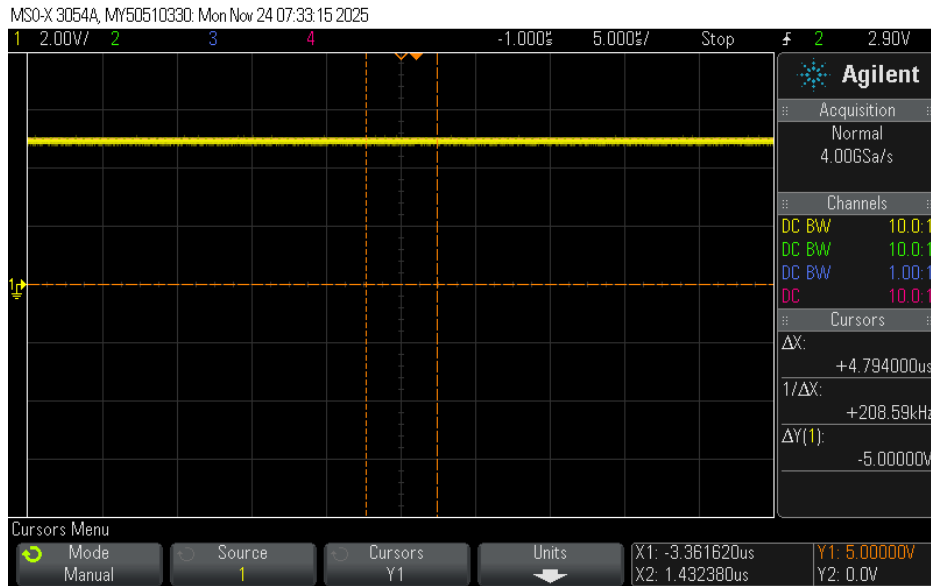


Figure 9: The yellow line is the stable, 5-volt output of the LM7805.

By using an op-amp (from an LM358 chip) to compare the 5-volt reference voltage with the output of the buck converter and then feeding that difference into the LM311, the duty cycle of the buck converter is automatically adjusted in order to ensure that the buck converter output stays at a steady 5 volts. After implementing feedback control, Figure 10 shows the final output of the buck converter with a 47Ω power resistor load, while Figure 11 shows the final output with a DC motor load. In both cases, the buck maintains a stable 5-volt output with minimal noise.

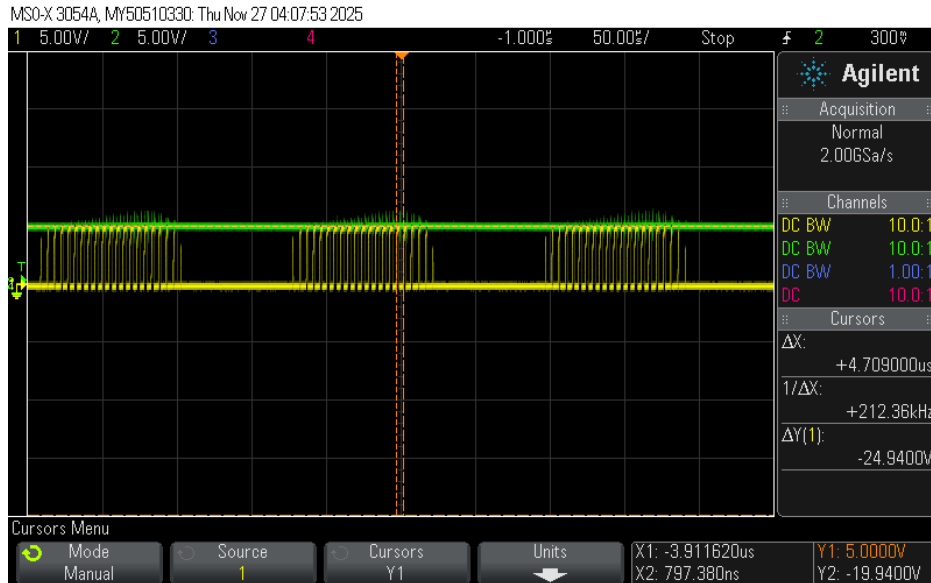


Figure 10: The yellow line is the LM311 output. The green line is the output of the buck converter with feedback when using a 47Ω power resistor load.

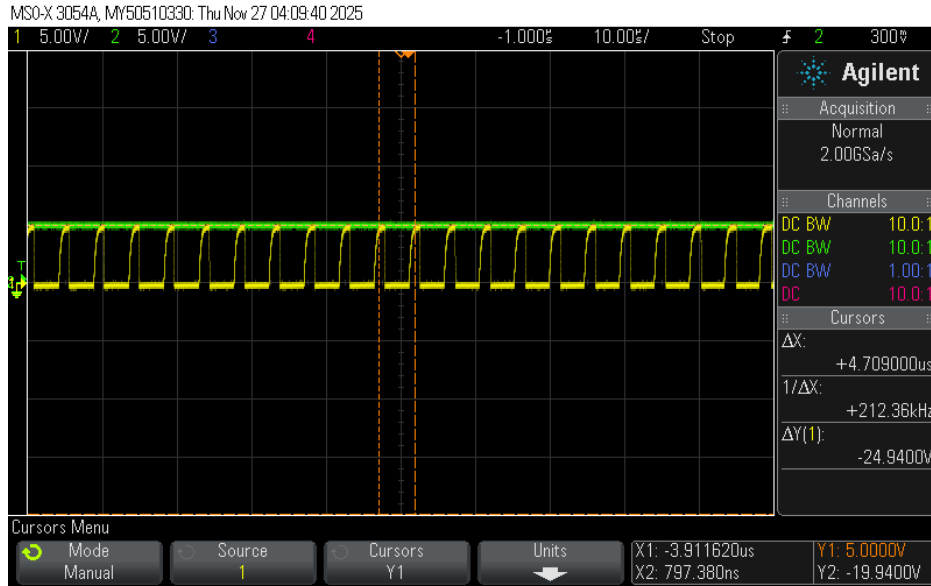


Figure 11: The yellow line is the LM311 output. The green line is the output of the buck converter with feedback when using a DC motor load.

4 DCM Boost Converter

The DCM boost converter went through multiple design iterations before eventually ending up as the following three smaller components: the combination of a roughly 10-2000 Hz clock ramp and a 200 kHz clock ramp, a push-pull gate driver, and the boost converter itself.

The first part is a 10-2000 Hz square wave. By using an LM555 with a 100kΩ potentiometer, this square wave was made to have an adjustable frequency ranging from roughly 12 Hz to 3 kHz, with much finer precision allowed for the lower part of that range.

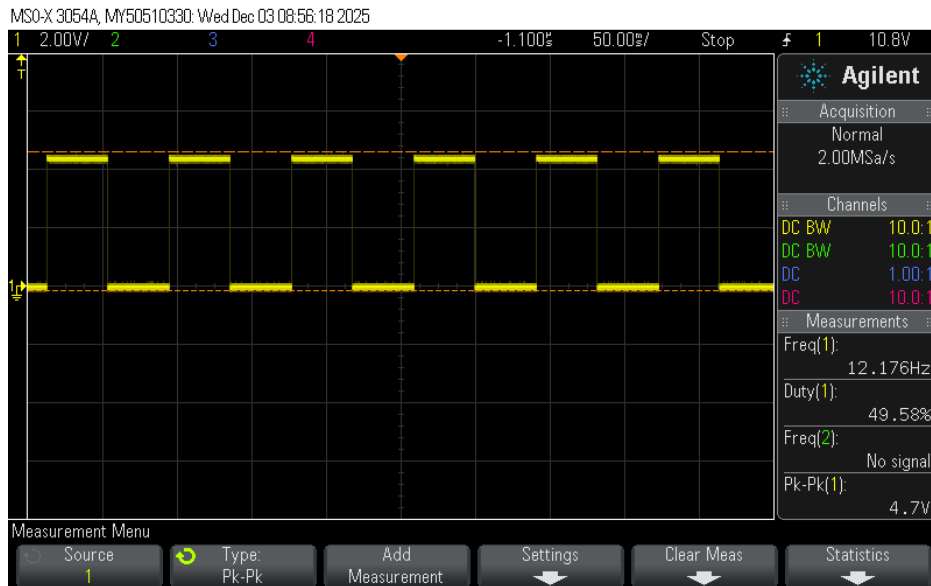


Figure 12: The yellow line is the LM311 output. The green line is the output of the buck converter with feedback when using a DC motor load.

However, if the boost converter were to run at such low frequencies, that would require enormous L and C

values. To solve this problem, the low-frequency square wave is combined with a higher-frequency sawtooth wave. Essentially, when the low-frequency square wave is low, then the circuit shuts off. But when the low-frequency square wave is high, a 200 kHz square wave is sent to the push-pull gate driver, allowing the boost converter to operate as if it has a switching frequency of 200 kHz.

By feeding both the low-frequency square wave (after using a transistor and a 2.4-volt reference voltage to ensure that the low-frequency square wave has a peak at the center of the sawtooth wave) and the high-frequency sawtooth wave to the LM311, the LM311 is able to output the desired wave. Figure 13 shows the inputs given to the LM311, while Figure 14 shows the corresponding output of the LM311.

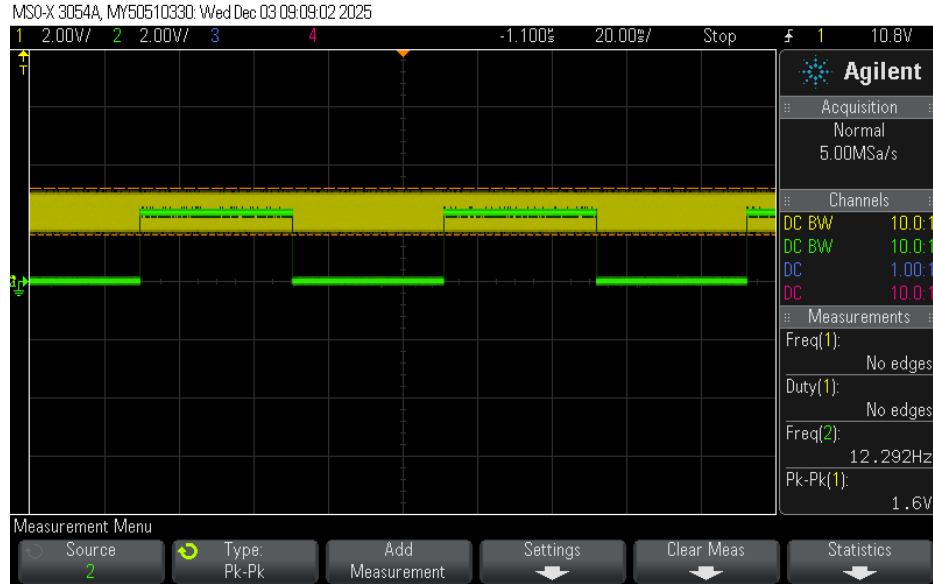


Figure 13: The yellow line is the high-frequency sawtooth wave given to pin 3 of the LM311. The green line is the low-frequency square wave resized by a transistor and given to pin 2 of the LM311.

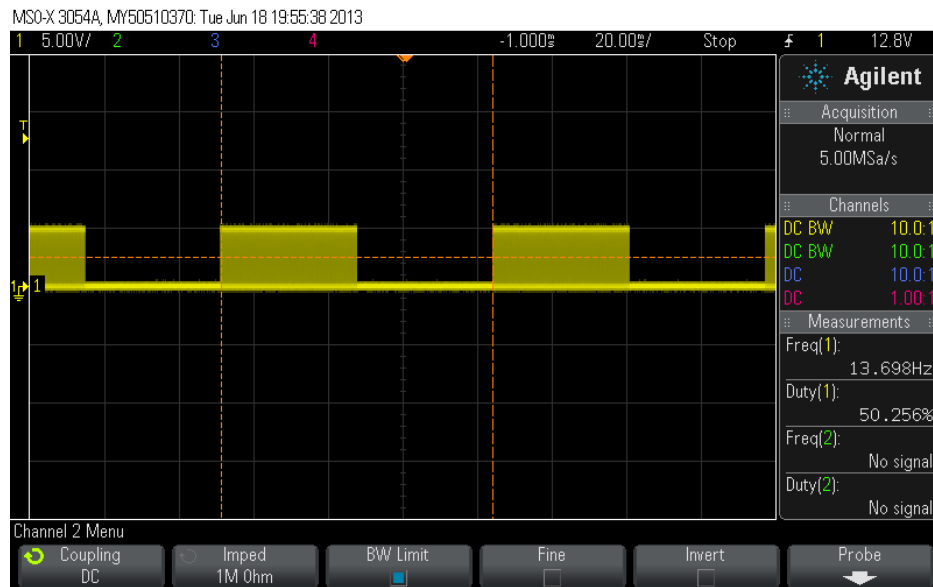


Figure 14: The yellow line is the output of the LM311.

Next, the output of the LM311 is given to the push-pull gate driver, which consists of two transistors that turn the logic-level MOSFET on and off as needed. Note that using a logic-level MOSFET (specifically, a ZVN3306A)

allows the entire system to be powered by the 5-volt input, removing the necessity of any self-start circuitry.

This MOSFET is then used as the low-side switch for the boost converter. As shown in Figure 15, the DCM boost converter is then able to properly power the 12-volt LED array, allowing it to flash at frequencies ranging from roughly 12 Hz to 3 kHz.

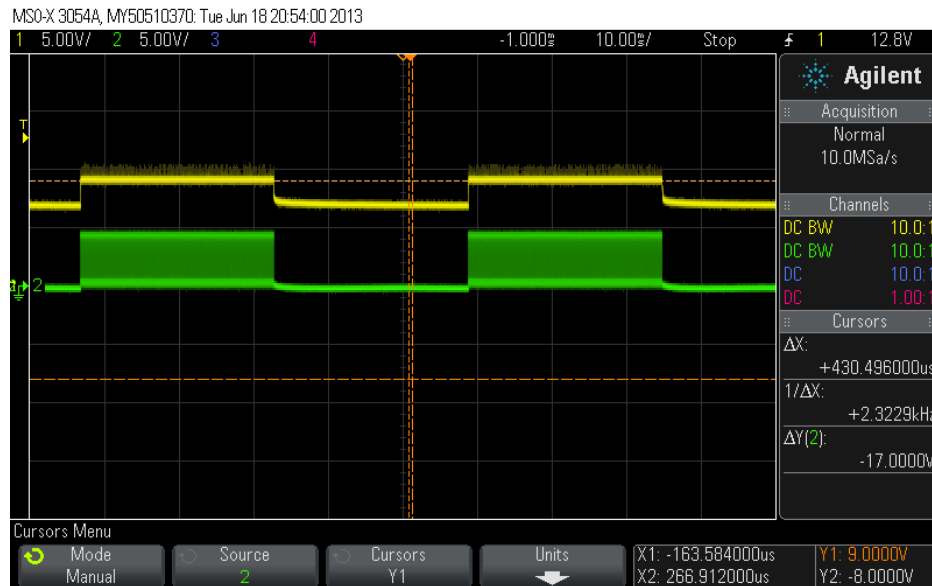


Figure 15: The yellow line is the input to the LED array (i.e. the load of the DCM boost converter). The green line is the input to the MOSFET gate.

However, effective LED stroboscopes operate with a very small duty cycle in order to produce sharp, clear images. Adding a diode and a 10kΩ resistor to the low-frequency square wave generator (specifically, using those components to connect pins 6 and 7 of the LM555) reduces the duty cycle to roughly 12%, although in future iterations, this value could be dropped even further to get rid of image blurriness. This change can be seen in Figure 16.

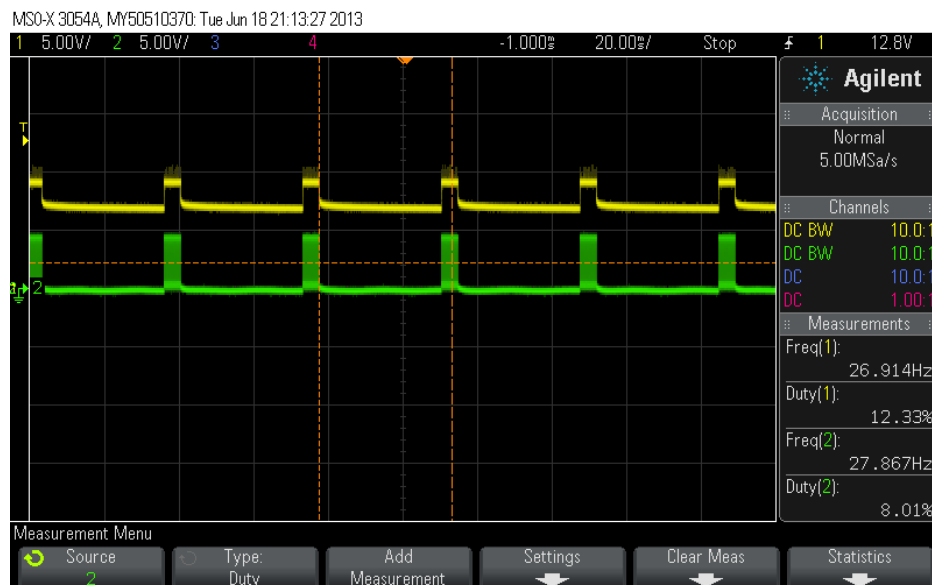


Figure 16: The yellow line is the input to the LED array (i.e. the load of the DCM boost converter). The green line is the input to the MOSFET gate.

5 Final LED Stroboscope

The end result of this project is a functioning LED stroboscope that demonstrates how changing the frequency of a flashing light can make a moving object appear to move in different speeds, different directions, or not even move at all.

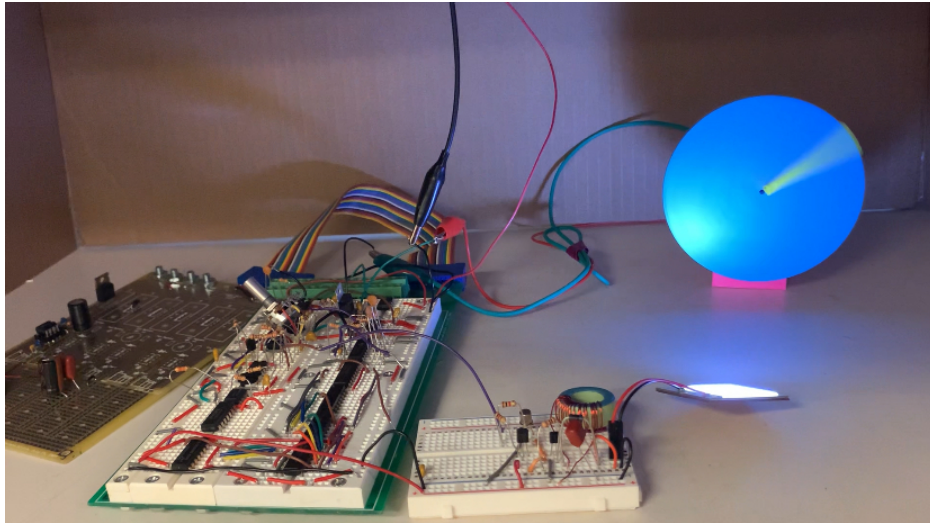


Figure 17: Final LED stroboscope design!